

NAMSSKOGAN, Norway

WMO: 012810

Lat: 64.7419N

Lon: 12.8458E

Elev: 459

StdP: 14.45

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.4	(c) -3.1	(d) -14.5	(e) 2.5	(f) -9.1	(g) -8.2	(h) 3.6	(i) -2.8	(j) 22.1	(k) 23.6	(l) 19.4	(m) 24.5	(n) 3.2	(o) 40	(p) 0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.5	(c) 80.0	(d) 62.0	(e) 75.4	(f) 60.4	(g) 71.3	(h) 58.3	(i) 64.0	(j) 76.8	(k) 61.7	(l) 72.9	(m) 59.5	(n) 69.2	(o) 6.6	(p) 100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.0	75.8	68.3	56.5	69.3	66.3	54.6	64.4	63.8	29.4	77.0	27.7	73.1	26.3	69.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 16.7	(b) 14.3	(c) 12.3	(e) -15.0	(f) 84.9	(g) 7.8	(h) 4.3	(i) -20.6	(j) 88.0	(k) -25.1	(l) 90.5	(m) -29.5	(n) 92.9	(o) -35.2	(p) 96.0		
(4) 16.7	(5) 14.3	(6) 12.3	(7) -15.4	(8) 67.0	(9) 7.7	(10) 2.8	(11) -20.9	(12) 69.0	(13) -25.4	(14) 70.7	(15) -29.8	(16) 72.3	(17) -35.4	(18) 74.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.7	22.0	22.6	27.8	36.2	45.1	52.5	58.9	55.8	48.6	38.6	29.8	24.9
	DBStd	15.13	11.39	11.01	9.17	5.70	7.18	6.52	6.51	5.51	5.41	6.06	8.95	10.37
	HDD50	4816	868	768	689	414	190	44	3	11	87	357	606	779
	HDD65	9649	1333	1188	1154	864	617	380	211	291	491	820	1056	1244
	CDD50	674	0	0	0	1	39	117	279	190	46	2	0	0
	CDD65	31	0	0	0	0	1	4	22	4	0	0	0	0
	CDH74	516	0	0	0	0	23	97	332	64	0	0	0	0
	CDH80	111	0	0	0	0	3	14	90	4	0	0	0	0
	WSAvg	5.7	6.5	6.6	6.2	5.5	6.0	5.6	5.3	5.1	5.3	5.4	5.5	5.7
	PrecAvg	50.8	5.4	4.5	4.4	3.1	2.3	2.7	3.5	3.5	4.9	5.6	4.8	5.9
(15) Precipitation	PrecMax	68.9	10.9	12.0	10.1	7.2	3.9	6.5	8.2	6.1	12.4	8.8	10.4	10.9
	PrecMin	33.3	1.1	0.2	0.3	1.0	0.7	0.0	1.3	1.4	1.5	2.2	0.6	1.8
	PrecStd	8.2	2.7	2.8	2.8	1.6	0.8	1.5	1.6	1.3	2.4	1.9	2.8	3.0
	PrecStd	8.2	2.7	2.8	2.8	1.6	0.8	1.5	1.6	1.3	2.4	1.9	2.8	3.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	42.4	48.6	62.4	76.7	82.3	87.9	80.1	70.1	57.2	48.4	43.9
		MCWB	39.2	38.7	39.3	49.0	58.2	61.2	66.6	63.6	57.6	50.3	46.4	40.7
	2%	DB	39.1	40.5	43.5	55.0	71.0	76.6	82.0	75.3	64.8	53.4	45.2	40.8
		MCWB	36.7	37.6	37.9	44.4	55.3	58.4	64.1	61.4	55.0	47.9	42.6	38.0
	5%	DB	36.7	38.2	40.7	50.1	65.4	70.8	77.4	71.3	61.0	50.0	42.0	38.5
		MCWB	34.4	35.5	36.9	41.7	52.6	56.6	62.3	59.6	53.6	46.4	39.0	35.9
	10%	DB	34.7	35.8	38.8	45.6	59.2	66.1	72.7	67.4	58.2	47.0	39.6	36.6
		MCWB	32.7	33.3	35.6	39.4	49.3	54.5	59.8	57.9	52.0	43.9	37.1	34.4
	0.4%	WB	39.9	39.6	41.0	49.2	59.3	63.1	69.3	65.1	58.5	50.9	47.1	41.4
		MCDB	41.0	41.5	43.8	61.2	75.4	78.6	83.1	76.7	66.1	55.4	48.3	42.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.2	37.8	39.0	45.5	56.2	60.2	65.6	62.5	56.5	48.4	43.0	38.5
		MCDB	38.8	39.9	42.4	54.1	68.8	72.9	78.2	73.0	63.5	52.6	45.3	40.2
	5%	WB	34.7	35.8	37.5	42.4	52.8	57.7	63.2	60.4	54.4	46.4	39.2	36.1
		MCDB	36.3	37.9	40.8	48.4	64.6	68.3	75.8	69.6	59.9	49.4	41.3	37.9
	10%	WB	32.9	33.6	35.6	40.1	49.6	55.2	60.8	58.4	52.4	44.3	37.3	34.5
		MCDB	34.4	35.6	38.4	44.7	58.1	65.2	71.0	66.3	57.2	46.7	39.3	36.3
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	12.1	16.2	16.1	18.1	17.8	18.5	18.1	15.0	10.8	9.0	11.1
		MCDBR	9.1	10.5	14.8	25.0	28.8	28.6	29.2	26.2	21.0	15.2	9.5	10.8
		MCWBR	7.7	8.3	10.5	15.6	16.0	14.1	13.8	13.8	13.1	11.1	7.9	9.0
	5% WB	MCDBR	8.7	10.3	13.4	21.3	26.9	25.0	25.9	22.7	19.2	12.5	9.4	10.5
		MCWBR	7.7	8.6	10.1	13.8	15.3	12.8	13.1	12.7	12.5	9.7	8.1	9.1
(40) Clear-Sky Solar Irradiance	taub		0.202	0.274	0.305	0.334	0.340	0.340	0.348	0.336	0.317	0.295	0.348	N/A
	taud		1.941	2.153	2.196	2.257	2.397	2.476	2.478	2.501	2.543	2.405	2.651	N/A
	Ebn at Noon		102	196	244	265	275	278	272	265	246	194	98	N/A
	Edh at Noon		8	18	28	33	32	30	29	26	20	15	7	N/A
	RadAvg		36	175	579	1153	1458	1502	1433	1101	602	246	60	13
(45) All-Sky Solar Radiation	RadStd		4	31	94	116	103	185	178	123	73	30	9	1

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-256	-320	N/A	N/A

Nomenclature: See separate page